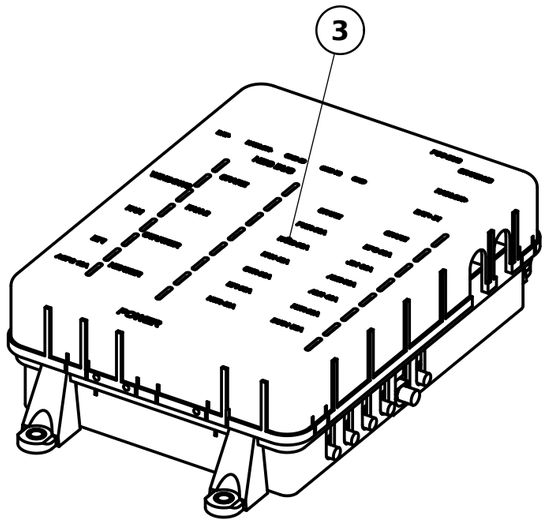
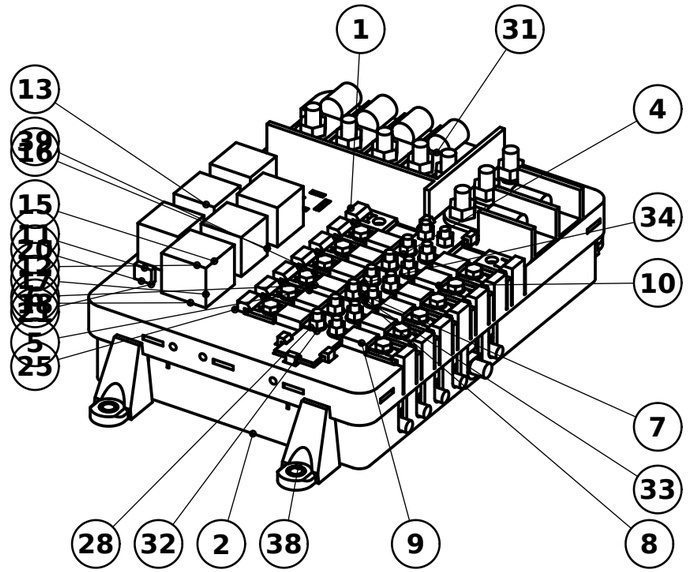
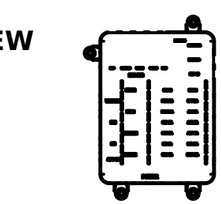
REVISIONS			
REV	DESCRIPTION	BY	DATE
R	#275: FB-101 base P	Claude	17/10/26
S	#277: FB-101 base R	Claude	17/10/26
T	#291: FB-101 base S, FB-102 pan F, FB-117 C-strap A	Claude	17/10/26
U	#297: FB-101 base T, FB-103 cover G, FB-104 R1 clip D	Claude	18/10/26
V	#303: FB-101 base U, FB-102 pan G, FB-104 R1 clip E	Claude	18/10/26
W	Buildability review: strap hook 40 deg lead-in, strap seat pocket (R181) - FB-101 base V; first assembly drawing	Claude	18/10/26

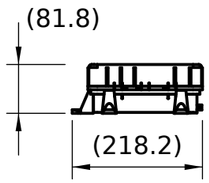


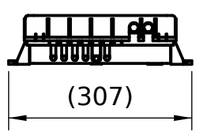
TRIMETRIC VIEW



TRIMETRIC - COVER REMOVED







FRONT / TOP / RIGHT VIEWS SCALE 1:10

ITEM	QTY	PART NUMBER	DESCRIPTION	MATERIAL / SPEC
1	1	ST185-FB-101	POWER BASE (WITH THE BUS ZONE)	PA12, 3D PRINTED
2	1	ST185-FB-102	POWER PAN (BELLY PAN)	PA12, 3D PRINTED
3	1	ST185-FB-103	POWER COVER	PA12, 3D PRINTED
4	2	ST185-FB-113/-114	POS BUS BAR L226 + NEG BAR L105	C110 Cu, TINNED
5	1	ST185-FB-117	OUTPUT C-STRAP 15 x 3 (14 FROM 1 BAR)	C110 Cu, TINNED
6	28	PEM CLSS-M5-2	SELF-CLINCHING NUT M5	C110 Cu, TINNED
7	6	AMI-30	AMI BOLT-DOWN FUSE 30 A	PURCHASED
8	3	AMI-40	AMI BOLT-DOWN FUSE 40 A	PURCHASED
9	1	AMI-50	AMI BOLT-DOWN FUSE 50 A	PURCHASED
10	2	AMI-60	AMI BOLT-DOWN FUSE 60 A	PURCHASED
11	1	ATM-10	ACPS MINI BLADE FUSE 10 A	PURCHASED
12	1	1-1414687-0	HEATER RELAY K_HTR, 1 FORM C, 12 V, 680 OHM	PURCHASED
13	3	1-1414147-0	MAXI ISO F7 RELAY 680 OHM (k_efi/k_str/k_fan)	PURCHASED
14	2	1-1393304-0	RELAY K_FAN2 / POWER (SEE NOTE 2) (NOT SHOWN)	PURCHASED
15	6	280756-4	MAXI RELAY LOAD RECEPTACLE 9.5 X 1.2	PURCHASED
16	4	280755-4	MAXI RELAY LOAD RECEPTACLE 9.5 X 1.2, 10-7 AWG	PURCHASED
17	12	42281-1	RELAY COIL RECEPTACLE 6.3 X 0.8, 18-14 AWG	PURCHASED
18	2	62998-2	HEATER RELAY LOAD RECEPTACLE 6.3 X 0.8	PURCHASED
19	1	42281-1	HEATER RELAY 87A RECEPTACLE 6.3 X 0.8 (SEE NOTE 2) (NOT SHOWN)	PURCHASED
20	1	1-968851-1	MINI FUSE RECEPTACLE MCP 2.8, 16 / 14 AWG	PURCHASED
21	1	1-968849-1	MINI FUSE RECEPTACLE MCP 2.8, 20 / 18 AWG	PURCHASED
22	1	RNF-100-1/4-0-STK	INSULATION BUILD-UP SLEEVE 1/4, 2 ENDS X 15 MM (NOT SHOWN)	PURCHASED
23	1	RNF-100-1/8-0-STK	INSULATION BUILD-UP SLEEVE 1/8 (NOT SHOWN)	PURCHASED
24	1	RNF-100-3/32-0-STK	INSULATION BUILD-UP SLEEVE 3/32 (NOT SHOWN)	PURCHASED
25	1	ST185-FB-104	R1 RETAINER CLIP	PA12, 3D PRINTED
26	4	94459A140	M3 HEAT-SET INSERT (OPTIONAL COVER BACKUP) (NOT SHOWN)	BRASS
27	4	SSCF-M3-10-A4	M3 x 10 CAP SCREW (OPTIONAL COVER BACKUP) (NOT SHOWN)	A4 STAINLESS
28	9	Klauke 2R5	TUBULAR LUG 10 MM2, M5 HOLE (8-10 AWG)	Cu, TINNED
29	1	Klauke 3R5MS	TUBULAR LUG 16 MM2, M5 HOLE (6 AWG) (NOT SHOWN)	Cu, TINNED
30	2	36161	RING LUG 12-10 AWG, M5 / #10 HOLE (NOT SHOWN)	Cu, TINNED
31	6	Klauke 6R8	TUBULAR LUG 50 MM2, M8 HOLE	Cu, TINNED
32	14	RS 287-3921	HEX BOLT M5 x 10, LUG (UP INTO THE BOTTOM PEM)	A2 STAINLESS
33	14	RS 263-2205	HEX BOLT M5 x 16 / M8 x 25, BAR STUDS	A2 STAINLESS
34	14	RS 264-6432	KEPS NUT M5 / HEX NUT M8, BAR STUDS	A2 STAINLESS
35	8	RS 190-276	HEX BOLT M8 X 25 (BUS STUDS: 3 POS + 5 N (SEE NOTE 2) (NOT SHOWN)	A2 STAINLESS
36	8	RS 530-769	HEX NUT M8 (BUS STUDS) (SEE NOTE 2) (NOT SHOWN)	A2 STAINLESS
37	8	RS 797-6068	PLAIN WASHER M8 (BUS STUDS) (NOT SHOWN)	A2 STAINLESS
38	4	EFC 10S00056	COMPRESSION LIMITER, FEET (4)	STEEL, ZINC
39	14	RS 287-3921	HEX BOLT M5 x 10, EAR (DOWN INTO THE TOP PEM)	A2 STAINLESS

NOTES:

1. OVERALL DIMENSIONS ONLY (REFERENCE). ITEMS MARKED (NOT SHOWN) ARE NOT MODELED; WIRES/HARNESS OMITTED FOR CLARITY.
2. RELAY / FASTENER ITEMS THAT SHARE ONE MODEL BODY ARE BALLOONED ON THE FIRST ITEM OF THE GROUP.
3. M5 BOLTS 4 N·m MAX (AMI SHEET). C-STRAP: PEM NUTS PRESSED BEFORE INSTALL, SEE FB-117.
4. C-STRAPS GO STRAIGHT DOWN INTO THEIR WELLS: THE LIPS PUSH THE SNAP HOOKS OUT (40 DEG LEAD-IN) AND THE HOOKS SNAP BACK OVER THE LIP ENDS (R181).
5. TERMINALS GO UP FROM UNDER THE DECK, CONTACT BOX FIRST, UNTIL THE LANCE CLICKS; PULL THE WIRE BACK LIGHTLY TO CHECK (TE 114-2025 3.6, 114-20021 3.7).
6. DO NOT TIE THE HARNESS CLOSER THAN 38 mm TO THE DECK UNDERSIDE, AND DO NOT DRESS WIRES SHARPLY TO ONE SIDE (TE 114-20021 3.7 C).

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
BREAK SHARP EDGES 0.2-0.4
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	W	SEE BOM

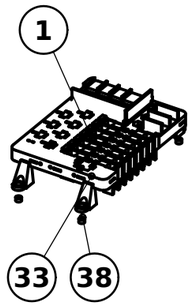
SCALE	UNITS	SIZE	SHEET	DATE	DRAWN
1:10	mm	B (11x17)	1 OF 2	2026-10-18	SteveCAD

SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

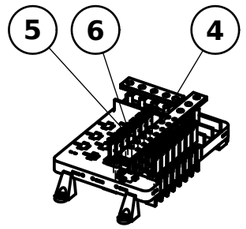
STEP 1: LIMITERS; BAR BOLTS FROM BELOW

EXPLODED TRIMETRIC (NOT TO SCALE)



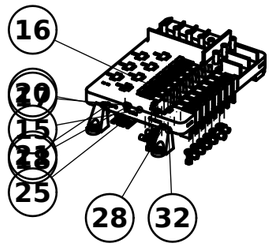
STEP 2: BARS; C-STRAPS STRAIGHT DOWN (NOTE 4)

EXPLODED TRIMETRIC (NOT TO SCALE)



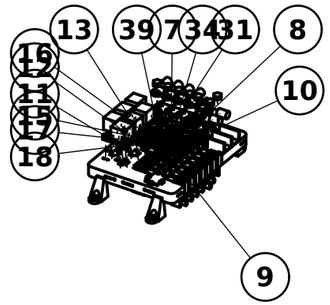
STEP 3: TERMINALS, R1 CLIP, LUGS UNDER THE STRAPS

EXPLODED TRIMETRIC (NOT TO SCALE)



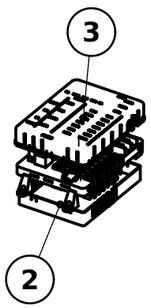
STEP 4: FUSES, EAR BOLTS, NUTS, LUGS, RELAYS

EXPLODED TRIMETRIC (NOT TO SCALE)



STEP 5: BELLY PAN + COVER

EXPLODED TRIMETRIC (NOT TO SCALE)



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
BREAK SHARP EDGES 0.2-0.4
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER FUSE BOX - ASSEMBLY

PART / DWG NO.	REV	MATERIAL
ST185-FB-100P	**W**	SEE BOM

SCALE	UNITS	SIZE	SHEET	DATE	DRAWN
AS SHOWN	mm	B (11x17)	2 OF 2	2026-10-18	SteveCAD

SOURCE
C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE